```
Sentiment columns: ['timestamp', 'value', 'classification', 'date']

Trades columns: Index(['account', 'coin', 'execution price', 'size tokens', 'size usd', 'side',
       'timestamp ist', 'start position', 'direction', 'closed pnl',
       'transaction hash', 'order id', 'crossed', 'fee', 'trade id',
       'timestamp'],
      dtype='object')

Sentiment shape: (2644, 4)
Trades shape: (211224, 16)
=======================
Sentiment missing:
 timestamp         0
value             0
classification    0
date              0
dtype: int64

Trades missing:
 account             0
coin                0
execution price     0
size tokens         0
size usd            0
side                0
timestamp ist       0
start position      0
direction           0
closed pnl          0
transaction hash    0
order id            0
crossed             0
fee                 0
trade id            0
timestamp           0
dtype: int64

Sentiment duplicates: 0
Trades duplicates: 0

================
Trades date range: 2023-05-01 2025-05-01
Sentiment date range: 2018-02-01 2025-05-02
```

==========

Daily Pnl:

	account	date	daily_pnl
0	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-11	0.000000
1	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-17	0.000000
2	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-18	0.000000
3	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-22	-21227.000000
4	0x083384f897ee0f19899168e3b1bec365f52a9012	2024-11-26	1603.100000
...	...	...	...
2336	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-27	1709.194807
2337	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-28	4008.588908
2338	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-29	8561.771838
2339	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	2025-04-30	2520.773814
2340	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	2025-05-01	1364.022527

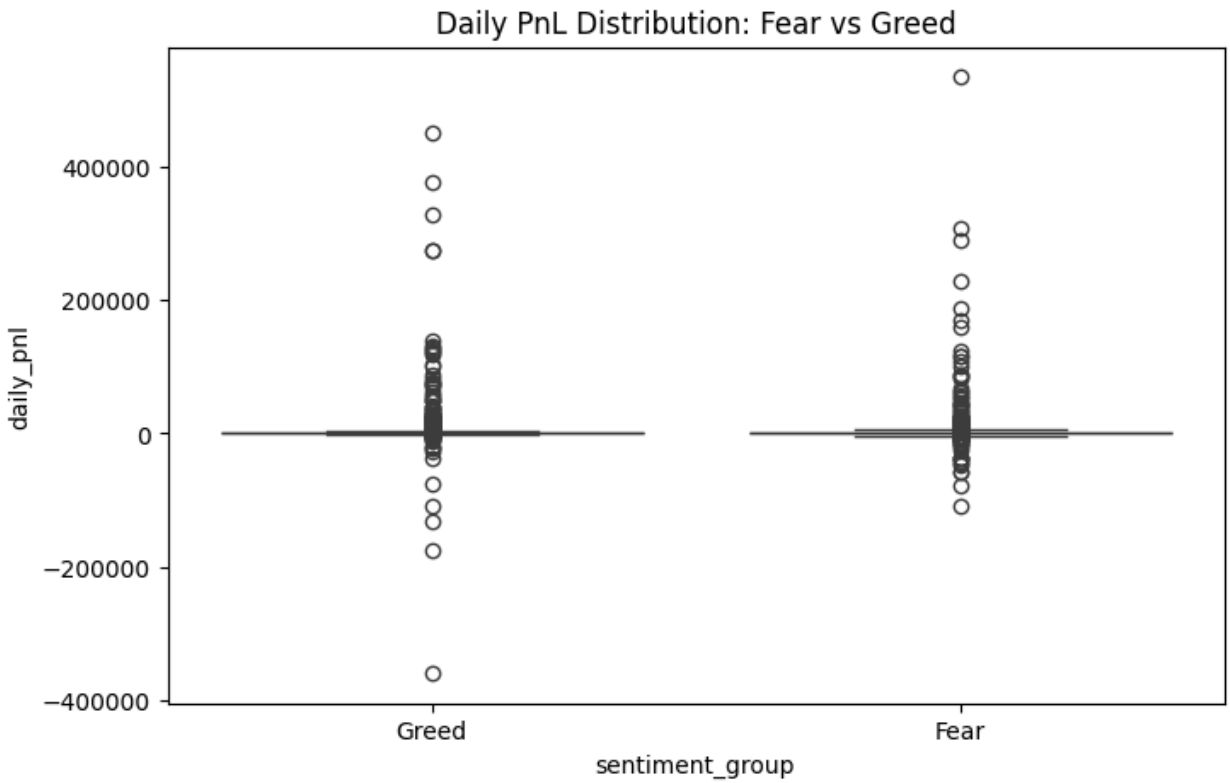
[2341 rows x 3 columns]

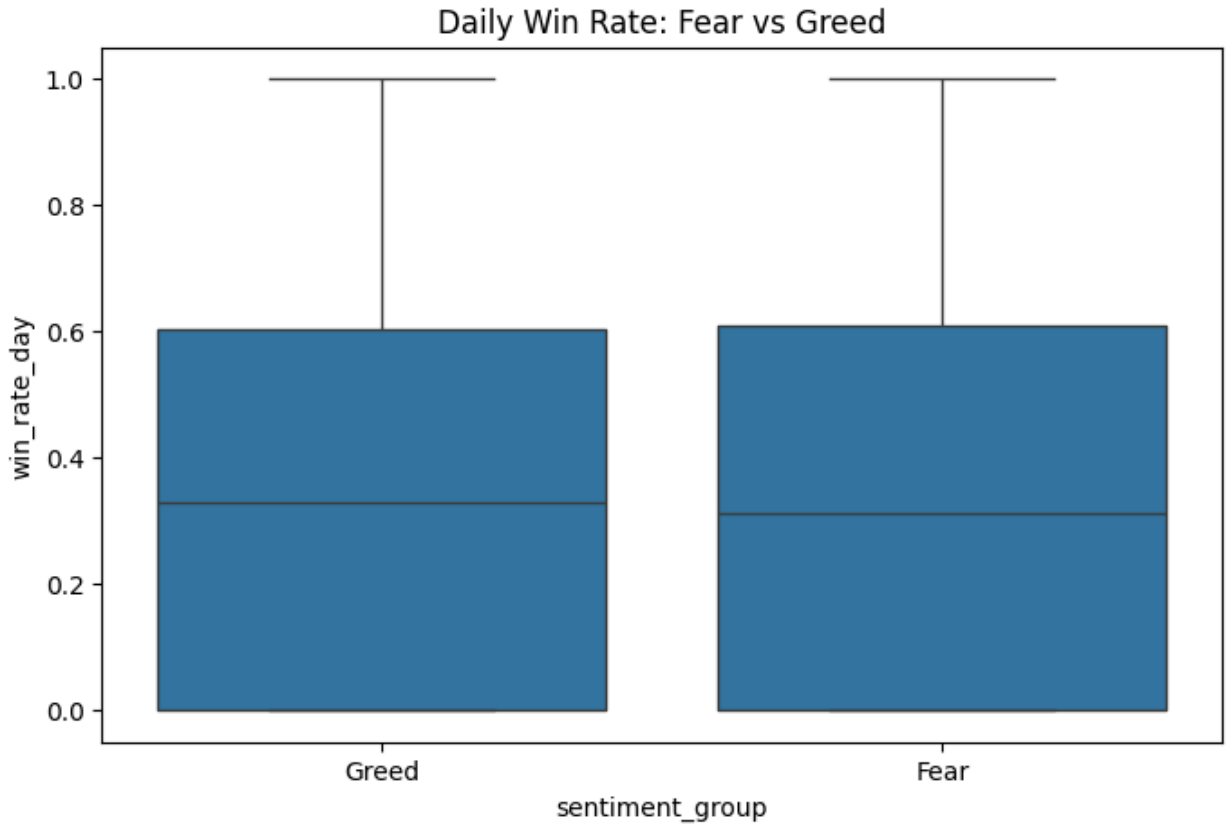
Win Rate:

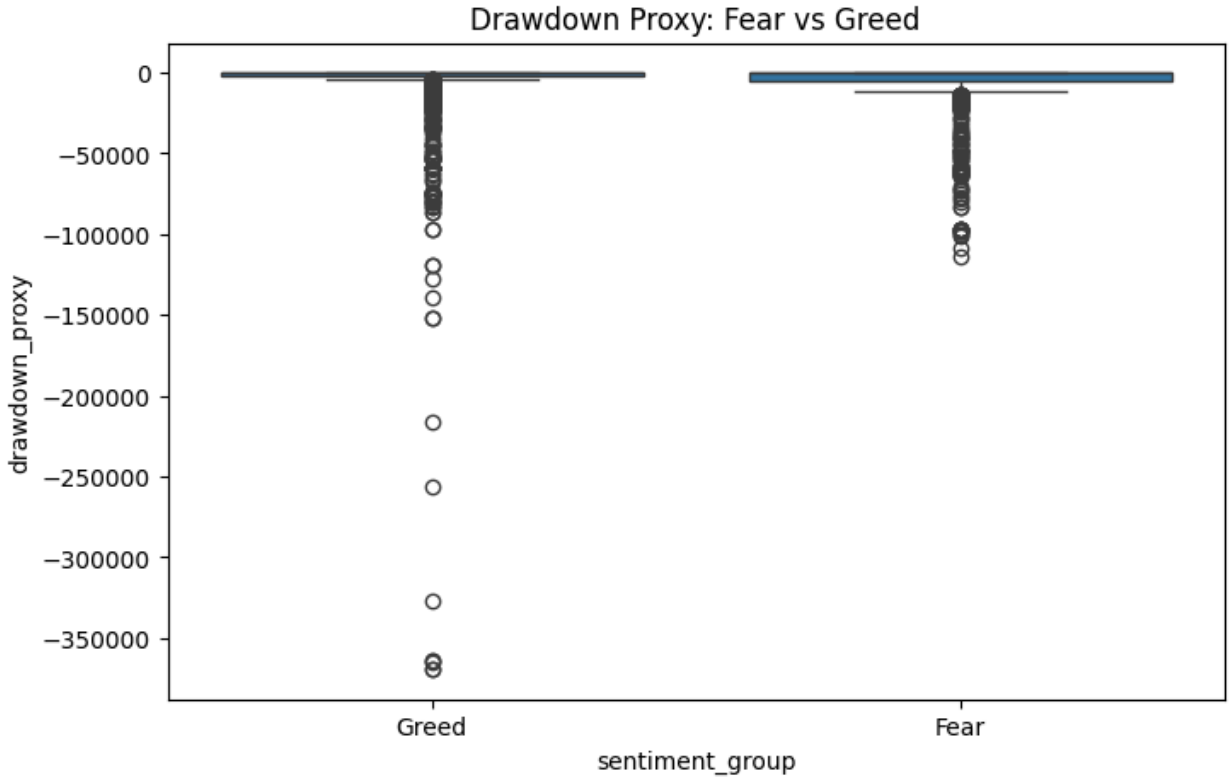
	account	win_rate
0	0x083384f897ee0f19899168e3b1bec365f52a9012	0.359612
1	0x23e7a7f8d14b550961925fbfdaa92f5d195ba5bd	0.442720
2	0x271b280974205ca63b716753467d5a371de622ab	0.301917
3	0x28736f43f1e871e6aa8b1148d38d4994275d72c4	0.438585
4	0x2c229d22b100a7beb69122eed721cee9b24011dd	0.519914
5	0x3998f134d6aaa2b6a5f723806d00fd2bbbbce891	0.455215
6	0x39cef799f8b69da1995852eea189df24eb5cae3c	0.327668
7	0x3f9a0aadc7f04a7c9d75dc1b5a6ddd6e36486cf6	0.475904
8	0x420ab45e0bd8863569a5efbb9c05d91f40624641	0.234987
9	0x430f09841d65beb3f27765503d0f850b8bce7713	0.484236
10	0x47add9a56df66b524d5e2c1993a43cde53b6ed85	0.353445
11	0x4acb90e786d897ecffb614dc822eb231b4ffb9f4	0.486226
12	0x4f93fead39b70a1824f981a54d4e55b278e9f760	0.360364
13	0x513b8629fe877bb581bf244e326a047b249c4ff1	0.401193
14	0x6d6a4b953f202f8df5bed40692e7fd865318264a	0.431795
15	0x72743ae2822edd658c0c50608fd7c5c501b2afbd	0.345912
16	0x72c6a4624e1dffa724e6d00d64ceae698af892a0	0.306294
17	0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	0.810876
18	0x7f4f299f74eec87806a830e3caa9afa5f2b9db8f	0.452213
19	0x8170715b3b381dffb7062c0298972d4727a0a63b	0.382743
20	0x8381e6d82f1affd39a336e143e081ef7620a3b7f	0.330194
21	0x8477e447846c758f5a675856001ea72298fd9cb5	0.261968
22	0x92f17e8d81a944691c10e753af1b1baae1a2cd0d	0.285387
23	0xa0feb3725a9335f49874d7cd8eaad6be45b27416	0.345787
24	0xa520ded057a32086c40e7dd6ed4eb8efb82c00e0	0.573141
25	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	0.408526
26	0xaf40fdc468c30116bd3307bcbf4a451a7ebf1deb	0.322097
27	0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	0.337134
28	0xb899e522b5715391ae1d4f137653e7906c5e2115	0.438611
29	0xbaaaf6571ab7d571043ff1e313a9609a10637864	0.467582
30	0xbd5fead7180a9c139fa51a103cb6a2ce86ddb5c3	0.327527
31	0xbee1707d6b44d4d52bfe19e41f8a828645437aab	0.428230

```python
print(trades_per_day)
```

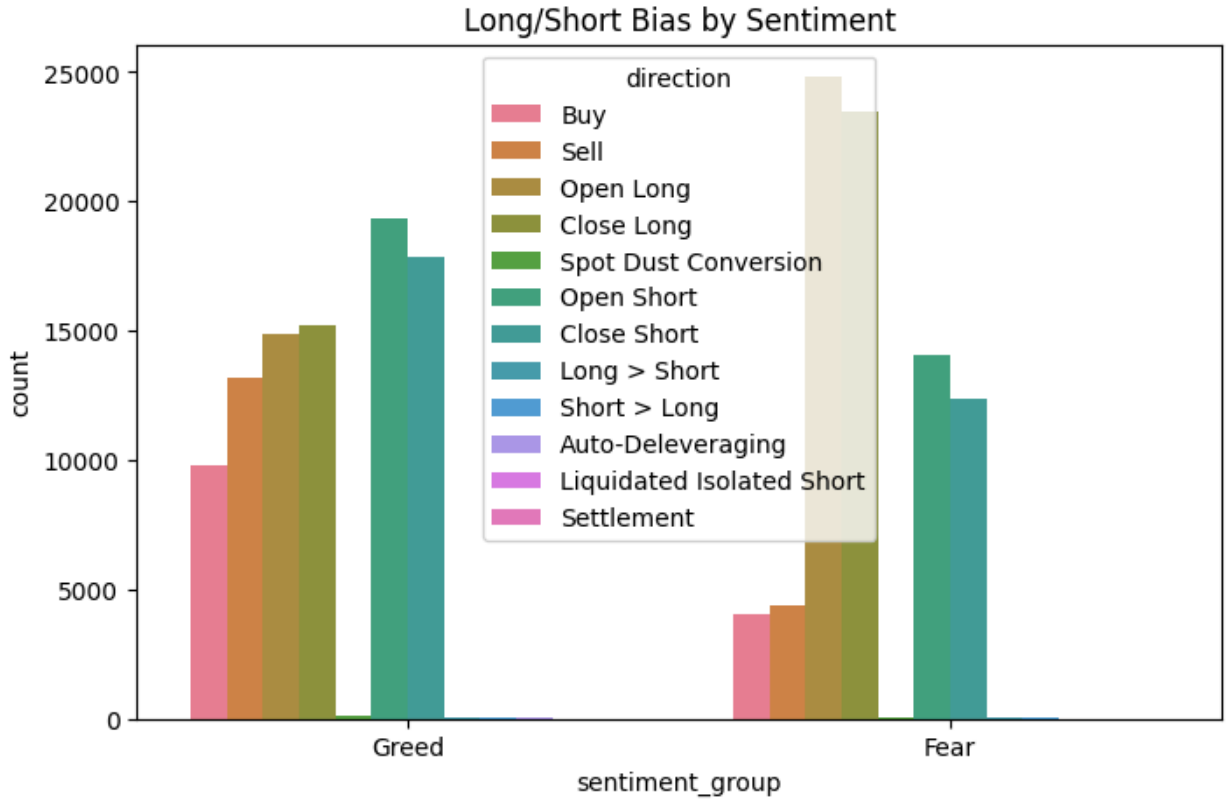
```
          date  num_trades
0   2023-05-01           3
1   2023-12-05           9
2   2023-12-14          11
3   2023-12-15           2
4   2023-12-16           3
..         ...         ...
475 2025-04-27         337
476 2025-04-28        1379
477 2025-04-29        2243
478 2025-04-30        1113
479 2025-05-01        1230

[480 rows x 2 columns]
```

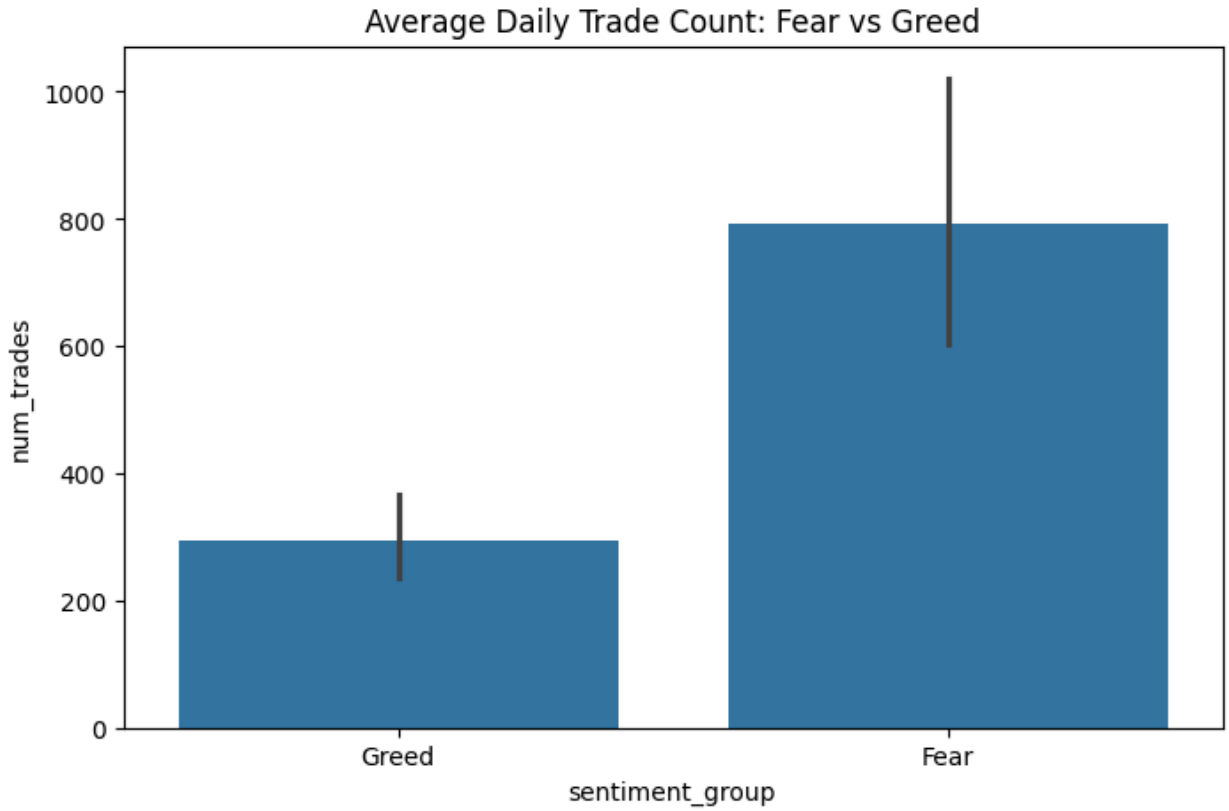


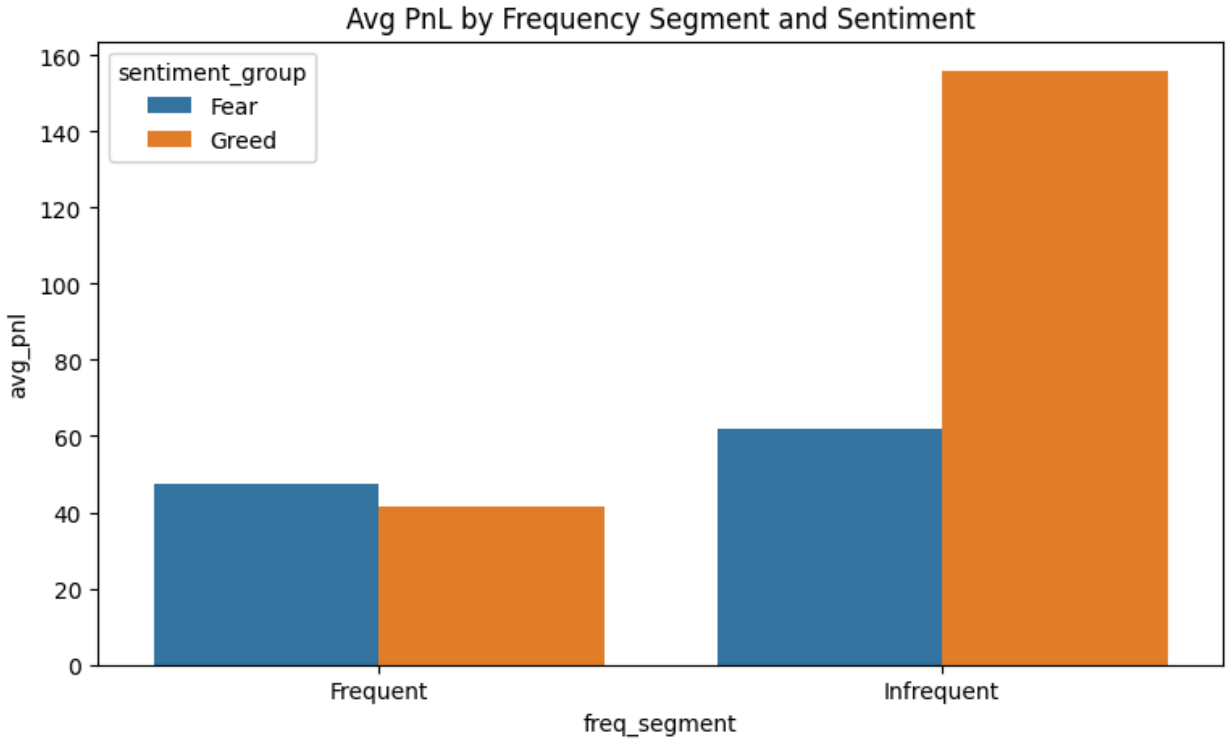

Daily Win Rate: Fear vs Greed
win_rate_day
1.0
0.8
0.6
0.4
0.2
0.0
Greed
Fear
sentiment_group

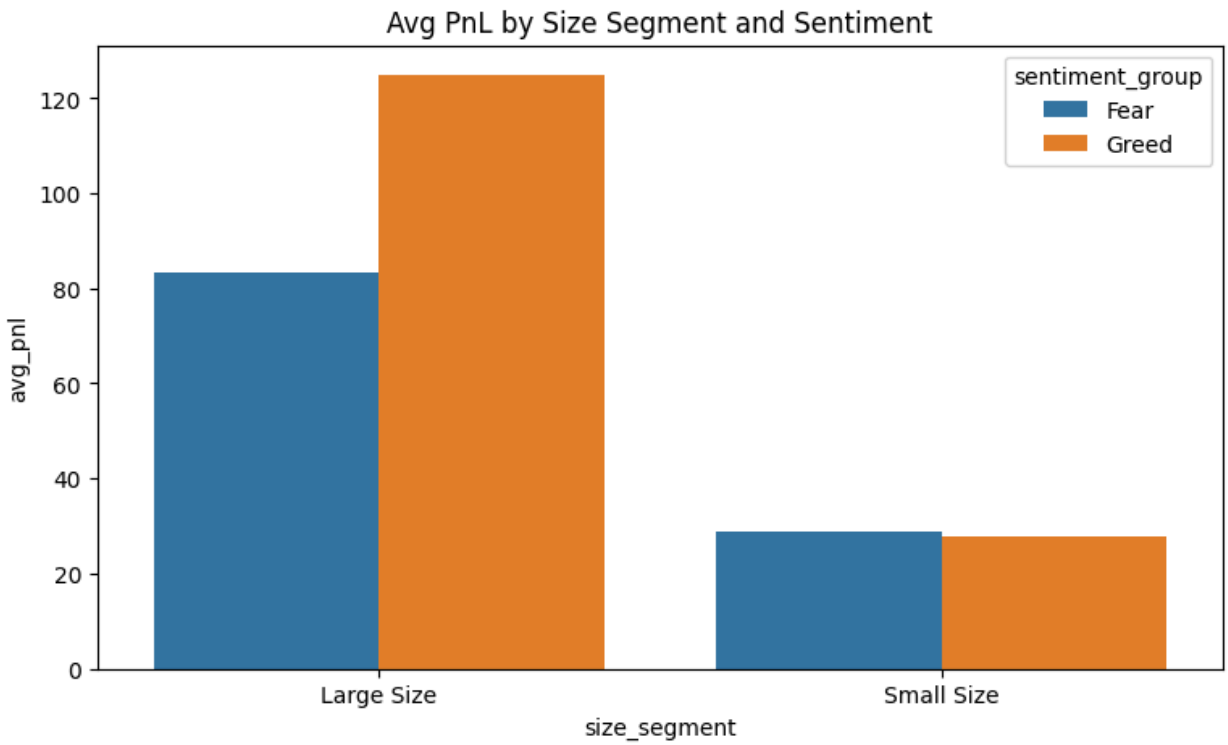

Drawdown Proxy: Fear vs Greed
drawdown_proxy
0
−50000
−100000
−150000
−200000
−250000
−300000
−350000
Greed
Fear
sentiment_group

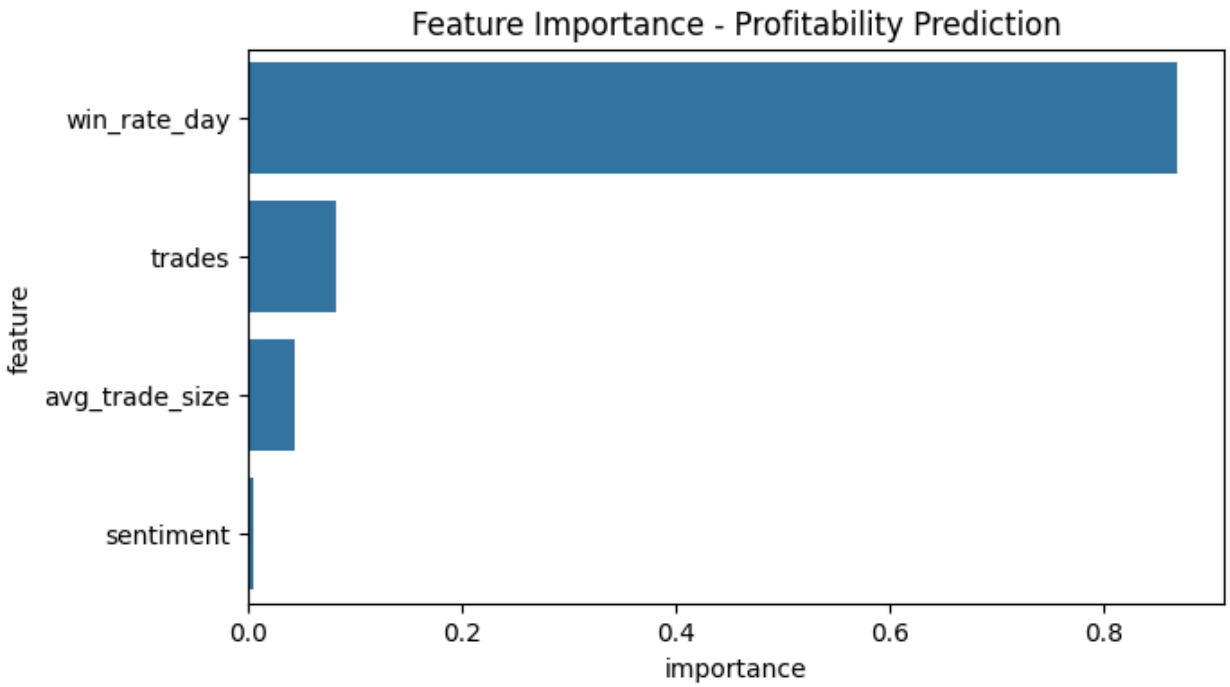
Trade Size Distribution: Fear vs Greed
size usd
10^6
10^5
10^4
10^3
10^2
Greed
Fear
sentiment_group


Long/Short Bias by Sentiment
count
25000
20000
15000
10000
5000
0
direction
Buy
Sell
Open Long
Close Long
Spot Dust Conversion
Open Short
Close Short
Long > Short
Short > Long
Auto-Deleveraging
Liquidated Isolated Short
Settlement
Greed
Fear
sentiment_group


Average Daily Trade Count: Fear vs Greed
num_trades
1000
800
600
400
200
0
Greed
Fear
sentiment_group


Avg PnL by Frequency Segment and Sentiment
sentiment_group
Fear
Greed
avg_pnl
160
140
120
100
80
60
40
20
0
Frequent
Infrequent
freq_segment



Model Accuracy:

```
Accuracy: 0.9423076923076923

Classification Report:
              precision    recall  f1-score   support

           0       0.96      0.88      0.92       175
           1       0.93      0.98      0.96       293

    accuracy                           0.94       468
   macro avg       0.95      0.93      0.94       468
weighted avg       0.94      0.94      0.94       468


Confusion Matrix:
 [[154  21]
 [  6 287]]
```